

Software Testing & Quality Assurance - Homework

Course Title : Software Testing & Quality Assurance

Course Code : CSE430

Section : 01

Topic: JUnit Testing (Java)

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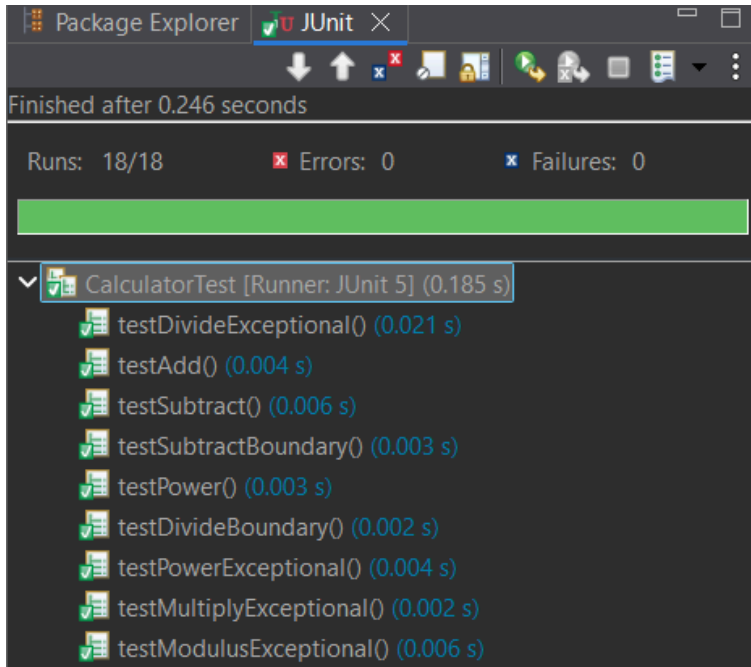
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Homework 1 - Enhanced Calculator

i) Screenshots



```

testAddBoundary() (0.003 s)
testSubtractExceptional() (0.005 s)
testMultiplyBoundary() (0.002 s)
testModulus() (0.002 s)
testPowerBoundary() (0.002 s)
testDivide() (0.002 s)
testMultiply() (0.003 s)
testAddExceptional() (0.005 s)
testModulusBoundary() (0.002 s)

```

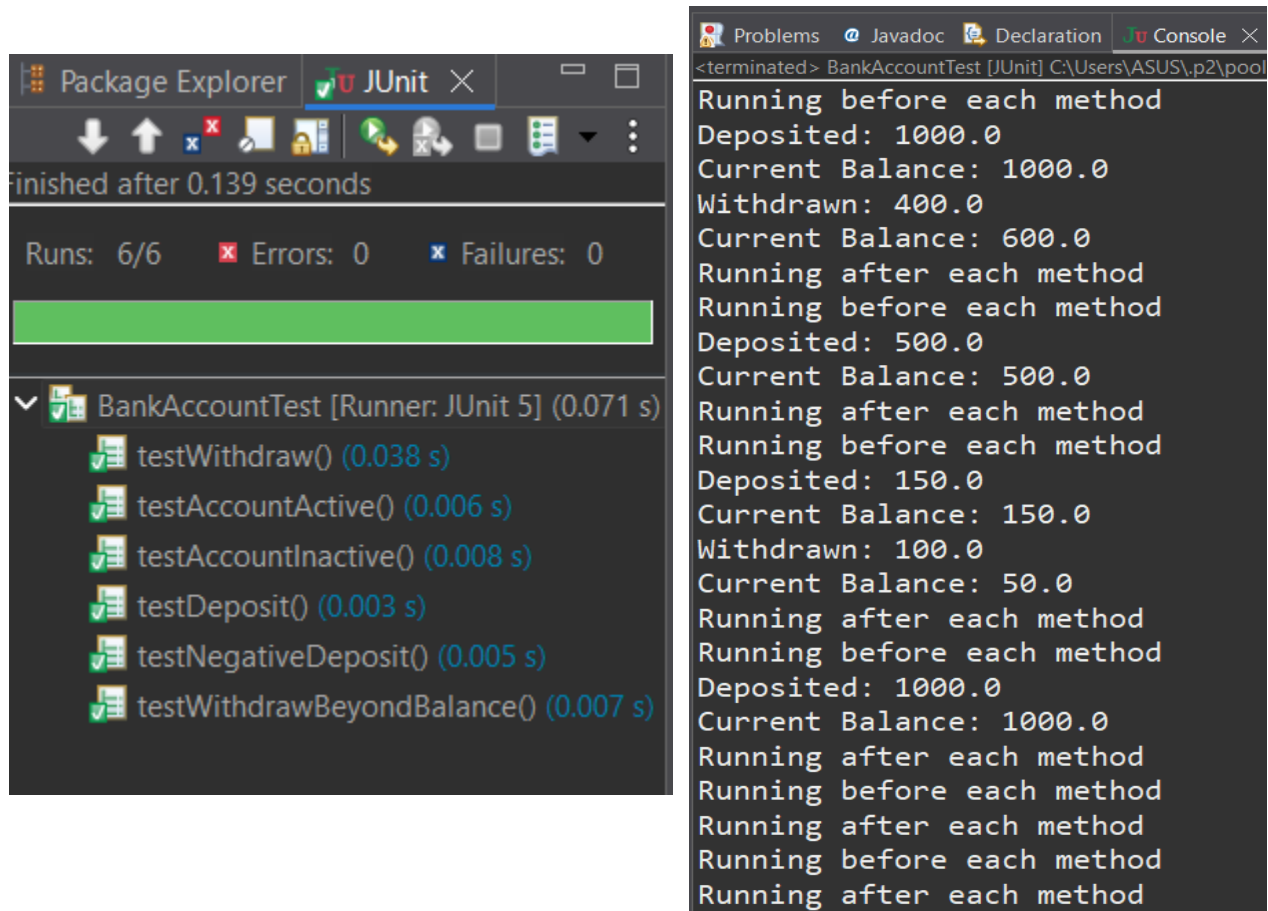
> BankAccountTest.java	0.0 %	0	105	105
✓ CalculatorTest.java	95.9 %	279	12	291
✓ CalculatorTest	95.9 %	279	12	291
▲ reset()	100.0 %	6	0	6
▲ setup()	100.0 %	8	0	8
▲ testAdd()	100.0 %	15	0	15
▲ testAddBoundary()	100.0 %	15	0	15
▲ testAddExceptional()	100.0 %	15	0	15
▲ testDivide()	100.0 %	15	0	15
▲ testDivideBoundary()	100.0 %	15	0	15
▲ testDivideExceptional()	100.0 %	11	0	11
▲ testModulus()	100.0 %	15	0	15
▲ testModulusBoundary()	100.0 %	15	0	15
▲ testModulusExceptional()	100.0 %	11	0	11
▲ testMultiply()	100.0 %	15	0	15
▲ testMultiplyBoundary()	100.0 %	15	0	15
▲ testMultiplyExceptional()	100.0 %	15	0	15
▲ testPower()	100.0 %	15	0	15
▲ testPowerBoundary()	100.0 %	15	0	15
▲ testPowerExceptional()	100.0 %	15	0	15
▲ testSubtract()	100.0 %	15	0	15
▲ testSubtractBoundary()	100.0 %	15	0	15
▲ testSubtractExceptional()	100.0 %	15	0	15
Calc	22.0 %	52	184	236

ii) Test-case table

Method	Input	Expected Output	Result
add()	(4,5)	9	Pass
add() Boundary	(5,0)	5	Pass
add() Exceptional	(2147483647,1)	-2147483648	Pass
subtract()	(9,5)	4	Pass
subtract() Boundary	(5,5)	0	Pass
subtract() Exceptional	(-2147483648,1)	2147483647	Pass
multiply()	(4,5)	20	Pass
multiply() Boundary	(4,0)	0	Pass
multiply() Exceptional	(2147483647,2)	-2	Pass
divide()	(20,5)	4	Pass
divide() Boundary	(20,1)	20	Pass
divide() Exceptional	(20,0)	IllegalArgumentException	Pass
power()	(2,3)	8	Pass
power() Boundary	(5,0)	1	Pass
power() Exceptional	(0,0)	1	Pass
modulus()	(10,3)	1	Pass
modulus() Boundary	(10,1)	0	Pass
modulus() Exceptional	(10,0)	IllegalArgumentException	Pass

Homework 2 – Account Validation

i) Screenshots



Package Explorer | JUnit

finished after 0.139 seconds

Runs: 6/6 Errors: 0 Failures: 0

BankAccountTest [Runner: JUnit 5] (0.071 s)

- testWithdraw() (0.038 s)
- testAccountActive() (0.006 s)
- testAccountInactive() (0.008 s)
- testDeposit() (0.003 s)
- testNegativeDeposit() (0.005 s)
- testWithdrawBeyondBalance() (0.007 s)

Console

```
<terminated> BankAccountTest [JUnit] C:\Users\ASUS\p2\pool
Running before each method
Deposited: 1000.0
Current Balance: 1000.0
Withdrawn: 400.0
Current Balance: 600.0
Running after each method
Running before each method
Deposited: 500.0
Current Balance: 500.0
Running after each method
Running before each method
Deposited: 150.0
Current Balance: 150.0
Withdrawn: 100.0
Current Balance: 50.0
Running after each method
Running before each method
Deposited: 1000.0
Current Balance: 1000.0
Running after each method
Running before each method
Running after each method
Running before each method
Running after each method
```

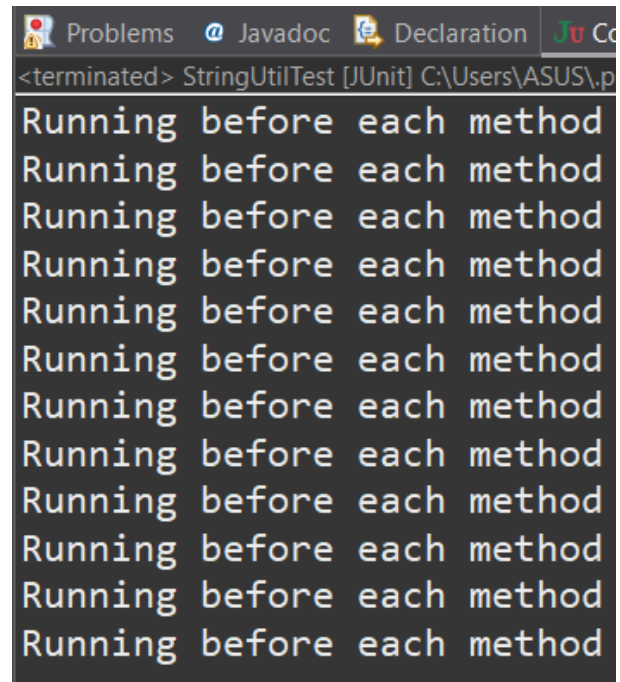
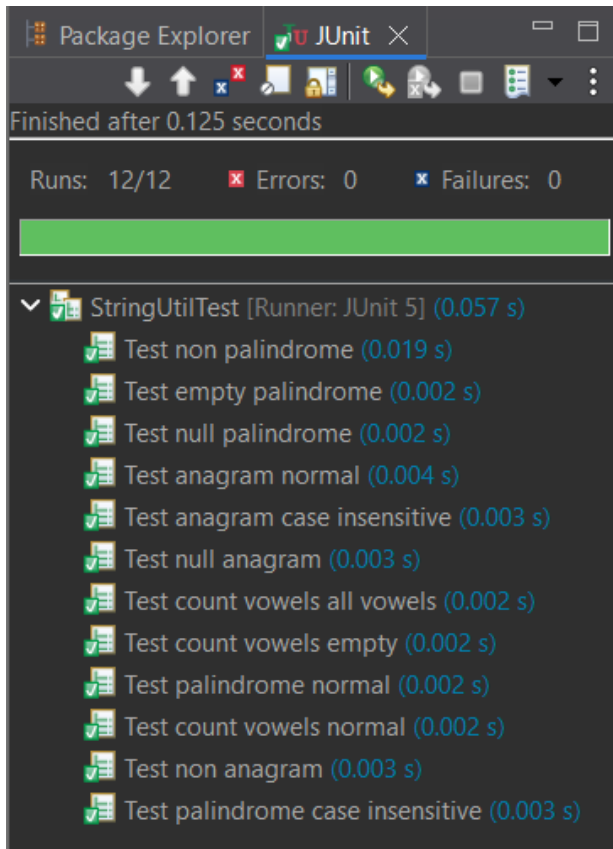
BankAccountTest (Jun 19, 2026 9:03:08 PM)					
Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions	
LabTask	19.8 %	164	666	830	
src	19.8 %	164	666	830	
CalcTest	16.0 %	95	499	594	
CalculatorTest.java	0.0 %	0	291	291	
StringUtilTest.java	0.0 %	0	198	198	
BankAccountTest.java	90.5 %	95	10	105	
BankAccountTest	90.5 %	95	10	105	
lastmethod()	100.0 %	7	0	7	
setup()	100.0 %	9	0	9	
testAccountActive()	100.0 %	11	0	11	
testAccountInactive()	100.0 %	15	0	15	
testDeposit()	100.0 %	17	0	17	
testNegativeDeposit()	100.0 %	6	0	6	
testWithdraw()	100.0 %	21	0	21	
testWithdrawBeyondBalance()	100.0 %	6	0	6	
Calc	29.2 %	69	167	236	

ii) Test - case table

Method	Input	Expected Output	Result
deposit()	1000	Balance = 1000	Pass
withdraw()	Deposit 1000, Withdraw 400	Balance = 600	Pass
withdraw() Overdraft	Withdraw 5000 from balance 0	IllegalStateException	Pass
deposit() Negative	Deposit -100	IllegalArgumentException	Pass
isActive() Valid	Deposit 500	true	Pass
isActive() Inactive	Deposit 150, Withdraw 100	false	Pass

Homework 3 - String Utility Testing

i) Screenshots



Problems Javadoc Declaration JUnit Console Coverage X

StringUtilTest (Jun 19, 2026 9:05:38 PM)

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
LabTask	35.8 %	297	533	830
src	35.8 %	297	533	830
CalcTest	33.3 %	198	396	594
CalculatorTest.java	0.0 %	0	291	291
BankAccountTest.java	0.0 %	0	105	105
StringUtilTest.java	100.0 %	198	0	198
StringUtilTest	100.0 %	198	0	198
setup()	100.0 %	9	0	9
testAnagram()	100.0 %	18	0	18
testAnagramCaseInsensitive()	100.0 %	18	0	18
testCountAllVowels()	100.0 %	13	0	13
testCountVowels()	100.0 %	13	0	13
testCountVowelsEmpty()	100.0 %	13	0	13
testEmptyPalindrome()	100.0 %	15	0	15
testNotAnagram()	100.0 %	18	0	18
testNotPalindrome()	100.0 %	15	0	15
testNullAnagram()	100.0 %	18	0	18
testNullPalindrome()	100.0 %	15	0	15
testPalindrome()	100.0 %	15	0	15
testPalindromeCaseInsensitive()	100.0 %	15	0	15
Calc	41.9 %	99	137	236

ii) Test - case table

Test Method	Input	Expected Output	Result
testPalindrome()	"madam"	true	Pass
testPalindromeCaseInsensitive()	"RaceCar"	true	Pass
testNotPalindrome()	"hello"	false	Pass
testNullPalindrome()	null	false	Pass
testEmptyPalindrome()	""	false	Pass
testCountVowels()	"hello"	2	Pass
testCountAllVowels()	"aeiou"	5	Pass
testCountVowelsEmpty()	""	0	Pass
testAnagram()	"listen", "silent"	true	Pass
testAnagramCaseInsensitive()	"Race", "Care"	true	Pass
testNotAnagram()	"hello", "world"	false	Pass
testNullAnagram()	null, "test"	false	Pass

iii) Observation (Edge Case)

One important edge case is handling strings with spaces and special characters. For example, "A man, a plan, a canal: Panama" should still return true as a palindrome after removing spaces and punctuation. Similarly, anagrams like "Dormitory" and "Dirty room" may require ignoring spaces. Adding such test cases would improve the reliability of the code.

Reflection paragraph

Through these three homework tasks, I learned a lot more about unit testing in Java and how JUnit works. Before this, I only had a basic idea of testing, but now I understand how to write test cases for normal, boundary, and exceptional situations. It helped me realize that testing is not only about checking correct outputs, but also about making sure the program handles invalid inputs properly. I also learned how annotations like `@BeforeEach`, `@AfterEach`, `@BeforeAll`, and `@AfterAll` work and when to use them. One thing I still did not fully understand is code coverage and how the percentage is calculated, so I would like to improve that part more. Overall, these tasks made me more confident in writing and organizing test cases in Java.